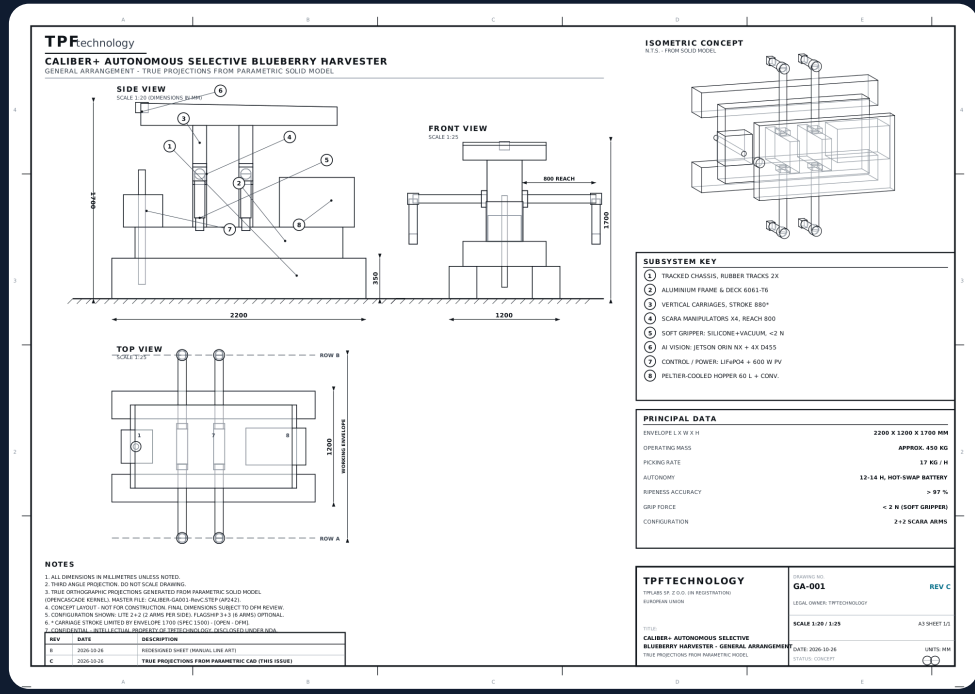
TPF LABS · AUTONOMOUS FIELD ROBOTICS

CALIBER+

The autonomous harvester that picks like hands, works like a machine

Selective harvesting of fresh-market blueberries. Two rows at once, berry by berry, zero crop loss, 12–14 hours of daily autonomy.

Investor deck · Seed round 2026



CALIBER+ · concept render · GA-001 Rev C

THE PROBLEM

Fresh-market berries are picked by hand, and hands are running out

💶 €2.4/kg and rising

Manual picking is the single largest cost in fresh-market blueberry production. Seasonal labour is scarce across the EU and getting scarcer every year.

🚜 Harvesters damage fruit

Over-the-row harvesters strip green and ripe berries together, bruise the crop and lose about 5% in the field — fruit drops out of the premium fresh category.

📉 Yield left in the field

Without reliable pickers, ripe berries wait days on the bush. Quality, weight and price all decay with every missed pass.

Premium fresh-market fruit needs **selective, gentle, round-the-clock picking** — exactly what humans do best and robots have never done. Until now.

THE PRODUCT

CALIBER+ — selective by design

🧠 AI ripeness vision

YOLOv9 + EfficientNet-B4 on NVIDIA Jetson Orin NX, trained on a 50,000-image berry dataset: finds every ripe berry, skips green ones, grades size.

>97% ripeness accuracy · ±1 mm grading

🫐 Soft hybrid gripper

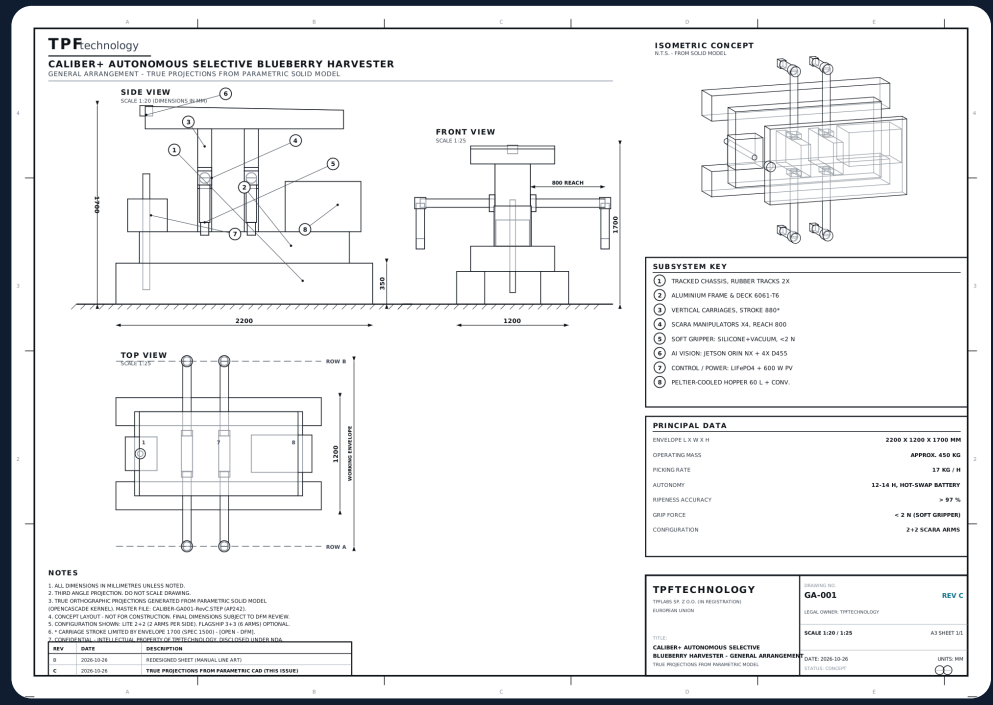
Silicone fingers with vacuum assist hold each berry with under 2 N of force — no bruising, no drops. Patent filed, FTO completed.

soft-touch ≤2 N · gripper BOM ≤€100

☀️ All-day autonomy

48 V $LiFePO_4$ with a 600 W solar roof and 45-second hot-swap. Tracked chassis for soft soil; LiDAR + RTK-GPS navigation, two rows at once.

12–14 h shifts · 4 SCARA arms · 17 kg/h



General arrangement GA-001 · concept layout

BUSINESS MODEL

Robot-as-a-Service: no capex for the farm

TIER	PRICE	WHAT THE FARM GETS
RaaS (per season)	**€26,000 / season**	≈€1.9/kg vs ≈€2.4/kg manual · service, software, updates · SLA: back in row ≤2 h
Purchase	**from €80,000**	Pilot-series unit, hybrid EU assembly, CE certification path

🧪 Pilot 30 / 60 / 90

Row survey & vision calibration → autonomous harvest on 1 ha with measured KPIs → full pilot report and season-contract option. No payment until a season contract is signed.

📊 Data as a product

Weekly harvest reports by ripeness class, yield maps and cost-per-kg — the analytics layer farms keep paying for.

🏭 Hybrid manufacturing

EU final assembly, optimized mechanics from an audited contract manufacturer (DFM package in review with WayKen, NDA signed).

TRACTION

De-risked step by step

✅ Gripper prototype: built

The core risk — gentle handling at speed — is lab-tested. Vision stack validated on a 50k-image dataset. Patent filed; freedom-to-operate analysis completed.

🏭 Supply chain engaged

Mutual NDA signed with contract manufacturer WayKen (CN); CALIBER+ tech package and GA-001 under DFM review for prototype and 50–100 units/year series.

🔬 Independent validation track

AgriFoodTEF route opened with ILVO (BE) — ripeness-model validation and 2027 field trial design — and a feasibility meeting with INRIA (FR) in progress.

🫐 First farm demand

GPO Polskie Jagody — one of Poland's largest blueberry producers — responded to outreach asking for machine details and cooperation models. Pilot talks started.

Early R&D is funded by **TPFtechnology**, the profitable parent development studio behind the project — founder capital, no debt.

ROADMAP & THE ASK

Seed round: €1.0–1.2M

from validated gripper to pilot fleet

2026 · Seed

Close the round · company TPFlabs Sp. z o.o. incorporation · engineering team scale-up · long-lead components for field prototype.

2027 · Field prototype & pilots

Field prototype built with contract manufacturing · 1–3 pilot farms run the 30/60/90 program · ILVO/AgriFoodTEF independent validation.

2028 · Commercial launch

Pilot-series production · CE under EU Machinery Regulation 2023/1230 · first RaaS season contracts · sales channel build-out.

Use of funds: engineering & prototype build (~45%), pilot fleet & field operations (~30%), certification & IP (~15%), team & operations (~10%).

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